

to maximum readable signal using VR5 (100-kilo-ohm). Moreover, LM358 protects the MCU from high signal voltages.

The program (GLCDscope.c) is developed and tested using AVR Studio 4 software. It can be compiled using AVR Studio 4 or above.

Once the circuit is ready, switch on the power. Set the fuse bits in the programming software and burn GLCDscope.hex file to ATmega16 through ISP port (CON3) using any AVR programmer.

Fuse bits. Set the fuse bits as follows for 16MHz crystal:

```
ckopt = on  sut0 = off sut1 = off
cksel0 = off cksel1 = off cksel2 = off
cksel3 = off
```

where off means programmed and on means unprogrammed. (Refer to Atmel datasheet for full details.)

Self-test and calibration. Initially, the GLCD shows text as shown in Fig. 2, followed by triangular patterns for a few seconds as self-test. It then starts reading the input ADC value continuously and displays the waveform along three horizontal lines—middle line being the zero-reference line, and top and bottom lines being reference lines of 50 per

cent of maximum positive and negative voltages of the input signal, respectively. In case the display is not clear, adjust VR1 for contrast adjustment of GLCD. Now, connect (short) both signal input pins. A horizontal line should appear near the middle (zero-reference) line. Adjust VR4 to match both signal horizontal and zero-reference lines by varying VR5.

Disconnect (open) signal

EFY Note
The source code
of this project
is available
for free download
at source.efymag.com

input lines and apply 5V DC to them. The signal horizontal line shifts up or down as per input 5V DC polarity. Adjust VR5 so that the signal horizontal line touches either top or bottom of GLCD screen. Repeat this process for 10V, 15V, 20V or as per the required voltage to be measured.

Construction and testing

An actual-size PCB layout of the simple GLCD scope is shown in Fig. 2 and its components layout in Fig. 3. Assemble the components on the PCB as per the circuit diagram.

Connect 50Hz or 60Hz AC signal from a step-down transformer (say, 6V AC) to signal input connector CON2. Set VR5 knob to about 10V range. Set VR2 and VR3 to minimum to get maximum samples per second. A sine wave will appear on the GLCD screen, as shown in Fig. 5.



Fig. 2: Initial display in GLCD

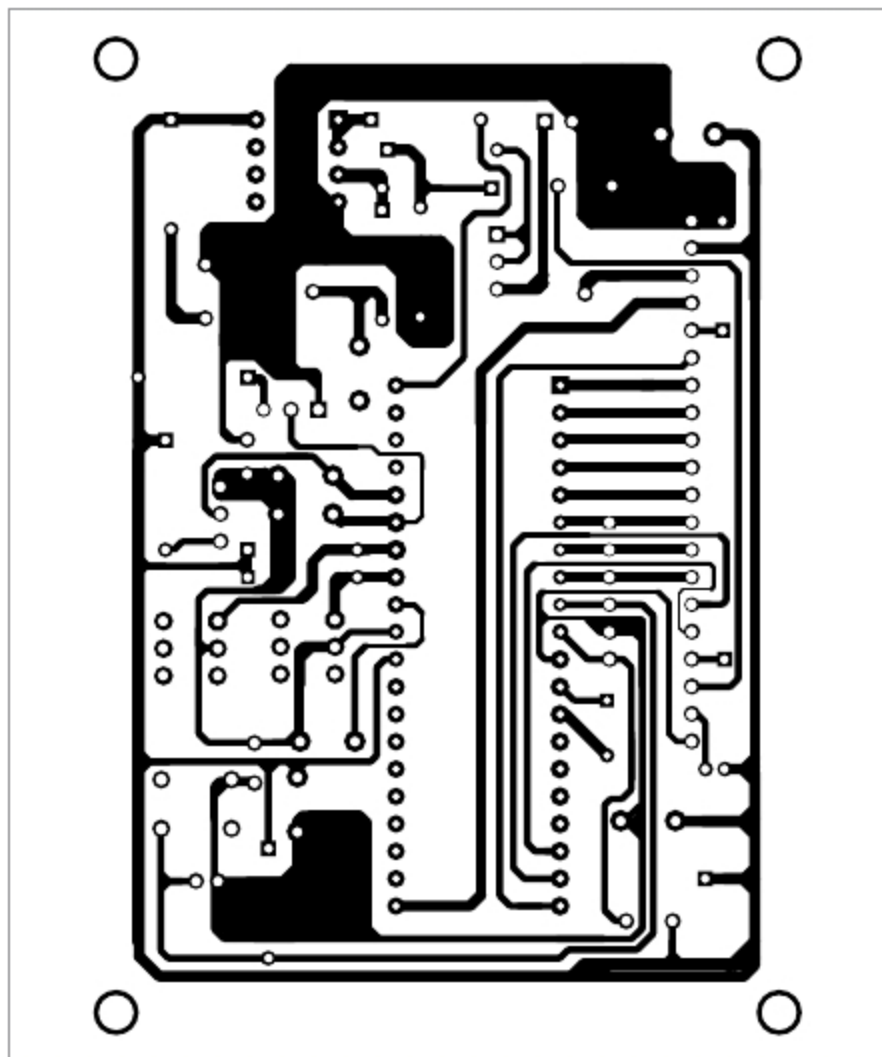


Fig. 3: Actual-size PCB layout of the GLCD scope

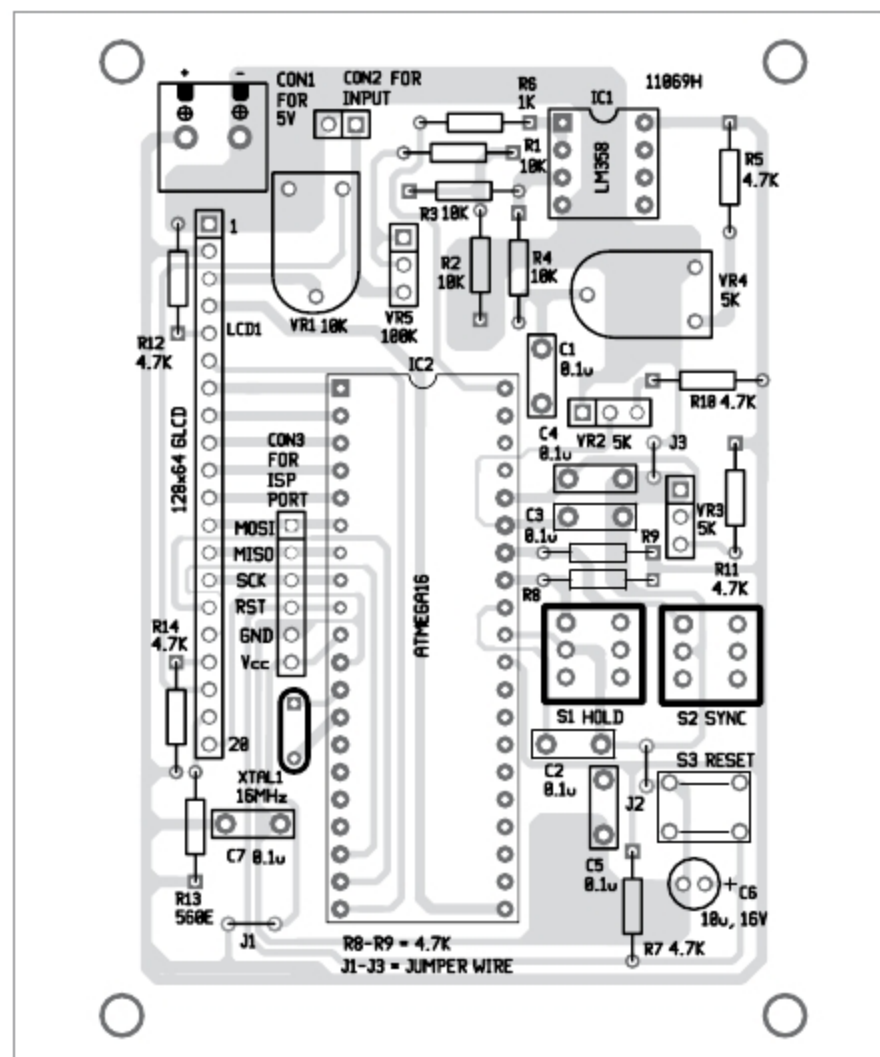


Fig. 4: Components layout for the PCB